Solution

Problems: Foundations

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Complex Numbers

$|c|$ - the magnitude of complex number c
$\arg\{c\}$ - the phase of the complex number c
c^* - the complex conjugate of c $\mathrm{Re}\{c\}$, $\mathrm{Real}\{c\}$ - the real part of c
$\mathrm{Im}\{c\}$, $\mathrm{Imag}\{c\}$ - the imaginary part of c

The complex number c may be written in rectangular coordinates as $c = \mathrm{Re}\{c\} + j\mathrm{Im}\{c\}$, and in polar coordinates as $c = |c|e^{j\arg\{c\}}$.

1. A complex number $a = 3 - 4j$ where j is the square root of -1. Find:

   a) $|a|$ $\quad |a| = \sqrt{3^2 + (-4)^2} = \sqrt{25} = 5$

   b) $\arg\{a\}$ in radians $\quad \arg\{a\} = \arctan\left(\frac{-4}{3}\right) \approx -.9273$ rads

   c) $\mathrm{Imag}\{a^*\}$ $\quad a^* = 3+4j$, $\mathrm{Imag}\{a^*\} = 4$

   d) aa^* $\quad (3-4j)(3+4j) = 3\cdot 3 - (j)^2\, 4\cdot 4 - 12j + 12j$
   $= 9 + 16 = 25$

2. Let $a = -5$. Find:

   a) $|a|$ $\quad |-5| = 5$

   b) $\arg\{a\}$ in radians $\quad -5 = 5e^{j\pi}$

   $\arg\{a\} = \pi$

3. Let $a = -4j$. Find:

a) $|a|$ $|-4j| = 4$

b) $\arg\{a\}$ in radians $-4j = 4e^{-j\pi/2}$

$\arg\{a\} = -\pi/2$

4. A complex function $a(t) = 2t - j2t^2$. Find $\arg\{a(t)\}$ in radians.

$\arg\{a(t)\} = \arctan\left(\frac{-2t^2}{2t}\right) = \arctan(-t)$

value in rads is a function of t

5. Let $a(t) = 5te^{j3t}$ and assume $t > 0$. What is $\mathrm{Re}\{a(t)\}$? Select all correct answers.

a) $5t$

b) $3t$

c) $5t\cos(3t)$

d) $5t\sin(3t)$

e) $5t\tan^{-1}(3t)$

f) $0.5a(t) + 0.5a^*(t)$

g) $0.5a(t) - 0.5a^*(t)$

h) $\sqrt{a(t)a^*(t)}$

$a(t) = 5te^{j3t}$

$= 5t(\cos(3t) + j\sin(3t))$

$\mathrm{Re}\{a(t)\} = 5t\cos(3t)$

Note $a(t) + a^*(t) = 2\mathrm{Re}\{a(t)\}$

6. Let $a(t) = e^{-t^2}e^{j2t}$ and $b(t) = |t|e^{j3t+\pi/2}$. Find $\arg\{c(t)\}$ if $c(t) = a(t)b(t)$. Select all correct answers.

$c(t) = |t|e^{-t^2}e^{j(2t+3t+\pi/2)}$

$= |c|\, e^{j\arg\{c\}}$

a) $-t^2 + 5t + \pi/2$

b) $5t + \pi/2$

c) $\pi/2$

d) $|t|e^{-t^2}$

e) $-t - \pi/2$

f) $\tan^{-1}\left\{\frac{3t+\pi/2}{2t}\right\}$

g) $\tan^{-1}\left\{\frac{\text{Im}\{b(t)\}}{\text{Re}\{a(t)\}}\right\}$

h) $t + \pi/2$

7. Write $x(t) = (1+2j)e^{j3t} + (1-2j)e^{-j3t}$ using real-valued sinusoids.

$x(t) = a(t) + a^*(t)$ if $a(t) = (1+2j)e^{j3t}$

$= 2\,\text{Re}\{a(t)\}$

$= 2\,\text{Re}\{(1+2j)(\cos 3t + j\sin 3t)\}$

$= 2\,\text{Re}\{\cos(3t) + 2j\cos(3t) + j\sin(3t) - 2\sin(3t)\}$

$= 2\cos(3t) - 4\sin(3t)$

a) $x(t) = \cos(3t) + 2\sin(3t)$

b) $x(t) = 2\cos(3t) + 4\sin(3t)$

c) $x(t) = \cos(3t) - 2\sin(3t)$

d) $x(t) = 2\cos(3t) - 4\sin(3t)$

e) $x(t) = 2\cos(3t) - 4j\sin(3t)$

f) $x(t) = 2\cos(3t) - 4\sin(6t)$

Can get the same answer using $e^{ja} = \cos(a) + j\sin(a)$ and multiplying / simplifying terms

Discrete- and Continuous-Time Frequency

8. A discrete-time sinusoid $x[n] = 3\cos(\pi n/5 + \pi/4)$ is obtained by sampling a continuous-time sinusoid $x(t)$ at sampling interval $T = 0.01$ s. That is, $x[n] = x(nT)$. Which of the following choices could have been the frequency of $x(t)$? Select all that are correct.

a) 100 Hz
b) 5 Hz
c) 10 Hz
d) 110 Hz
e) 90 Hz
f) 55 Hz

Recall discrete-time frequency is not unique. $\pi/5$ and $\pi/5 + \ell 2\pi$ give same sinusoids for all integer ℓ

Also $\omega = 2\pi f T$

use $\omega = \pi/5 + \ell 2\pi$ and $T = 0.01$

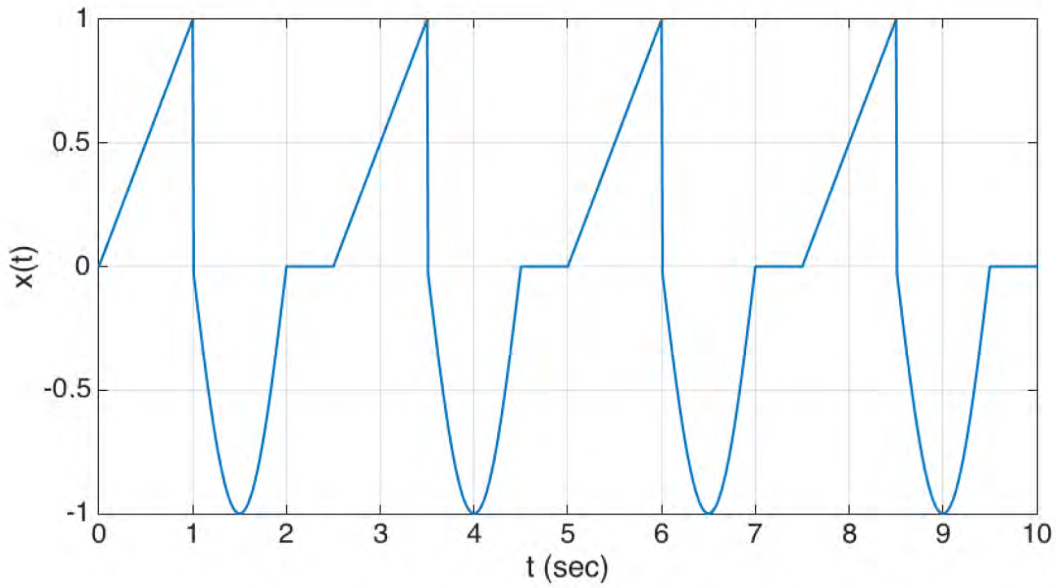
$\pi/5 + \ell 2\pi = 2\pi f (0.01)$

$100\left(\frac{1}{10} + \ell\right) = f$

$f = 10 + \ell 100$

Periodic Signals

9. Find the fundamental period in sec and fundamental frequency in Hz of the periodic continuous-time signal depicted below.



Graph shows 4 periods in 10 sec

$\Rightarrow$ fundamental period = 2.5 sec

fundamental frequency $= \frac{1}{2.5\,\text{sec}} = 0.4\,\text{Hz}$

10. If a signal $x(t)$ has fundamental period $T_x = 4$ s and $y(t)$ has fundamental period $T_y = 5$ s, what is the fundamental period of $x(t) + y(t)$?

11. If a signal $x(t)$ has period T_x s and $y(t)$ has period T_y s, what condition must be satisfied for $x(t) + y(t)$ to also be periodic? Assume m and k are both positive integers.

a) $T_x = T_y + m$

b) $T_x = mT_y$

c) $T_y = kT_x$

d) $mT_y = kT_x$

e) $T_xT_y = mk$

f) $T_x + T_y = m + k$

These are very similar problems. In 10. $x(t)$ has periods 4, 8, 12, 16, 20, 24... while $y(t)$ 5, 10, 15, 20, 25, ...

The smallest common period is 20 s, so this will be the period of the sum.

11. $x(t)$ and $y(t)$ must have a common period for the sum to be periodic.

Periods of $x(t)$: $T_x, 2T_x, 3T_x, \ldots \Rightarrow kT_x$

periods of $y(t)$: $T_y, 2T_y, 3T_y, \ldots \Rightarrow mT_y$

In order to have a common period, there must be integers k and m for which

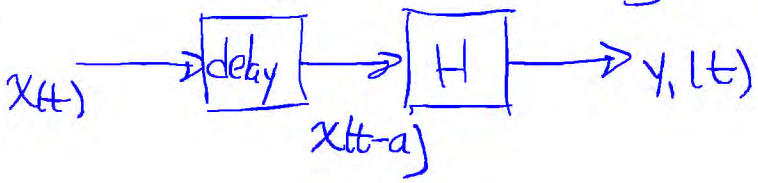
$$mT_y = kT_x$$

System Properties

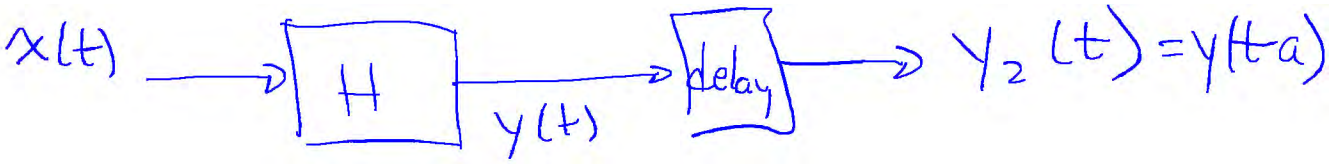
12. The output $y(t)$ of a system is obtained by delaying the input by 2 secs, multiplying by three, and changing its sign. That is, $y(t) = -3x(t-2)$.

a) Is this system time invariant? The behavior of this system: delay, scale by 3, invert, is independent of t. It always acts this way for all t.

Mathematically we change show this by checking whether



is the same as



To get $y_1(t)$, we delay $x(t-a)$ by 2, scale by 3, and invert: $y_1(t) = -3x(t-a-2)$

To get $y_2(t)$, we delay $y(t)$ by a

$y_2(t) = y(t-a) = -3x(t-2-a)$

So it doesn't matter whether I wait and then use the system ($y_1(t)$) or whether I use the system immediately ($y_2(t)$)

Time Invariant

b) Is this system memoryless?

No. Output at t depends on past value of the input

c) Is this system BIBO stable? yes ⇒ if $x(t)$ is bounded, $y(t)$ is also bounded. Formally, assume $|x(t)| \le M_x < \infty$ for all t.

$$\max_t |y(t)| = \max_t |-3x(t-2)| = 3M_x$$

so $|y(t)| \le 3M_x < \infty$ is bounded

d) Is this system linear? superposition?

If $x_1(t) \to y_1(t)$, $x_2(t) \to y_2(t)$, then does $H\{ax_1(t)+bx_2(t)\} = ay_1(t)+by_2(t)$?

$y_1(t) = -3x_1(t-2)$, $y_2(t) = -3x_2(t-2)$

$$H\{ax_1(t)+bx_2(t)\} = -3(ax_1(t-2)+bx_2(t-2))$$

$$= a\underbrace{[-3x_1(t-2)]}_{y_1(t)} + b\underbrace{[-3x_2(t-2)]}_{y_2(t)}$$

Linear

e) What is the impulse response $h(t)$ of the system?

i) can't be determined from information given

ii) $h(t) = -3u(t-2)$

iii) $h(t) = 3\delta(t-2)$

iv) $h(t) = 3u(t-2)$

v) $h(t) = -3\delta(t-2)$

vi) $h(t) = -3\delta(t+2)$

vii) $h(t) = 3\delta(t+2)$

$h(t) = H\{\delta(t)\}$

$= -3\,\delta(t-2)$

Impulse Response Descriptions

13. An LTI system has impulse response $h[n] = (1/2)^n u[n]$.

a) The input to the system is $x[n] = 2\delta[n-3]$. Use linearity and time invariance to find the output.

By time invariance $H\{\delta[n-3]\} = (1/2)^{n-3} u[n-3]$

by linearity $H\{2\delta[n-3]\} = 2(1/2)^{n-3} u[n-3]$

b) The input to the system is $x[n] = 4\delta[n] - 2\delta[n-3]$. Use linearity and time invariance to find the output.

$y[n] = H\{x[n]\} = 4h[n] - 2h[n-3]$

$y[n] = 4(1/2)^n u[n] - 2(1/2)^{n-3} u[n-3]$